

CarDiff — Research Spotlight

60-second speaking script • MIUA 2026

Target $\approx$ 60 s • $\approx$ 150 words • $\approx$ 150 wpm

Read one block per slide. **Bold** = light vocal stress. Advance the slide at each time marker.

Slide 1 • Motivation

0:00–0:19 • $\sim$ 47 w

Dental caries is the **most common disease in the world**, and catching it **early** on X-rays really matters. But AI for caries detection is held back by two problems: **too few annotated images**, and **severe class imbalance** — the early, superficial and medium lesions we most want to catch are the **rarest**.

Slide 2 • Technical Innovation

0:19–0:40 • $\sim$ 53 w

Our solution is **CarDiff** — a segmentation-guided **latent diffusion** model that synthesizes realistic dental radiographs directly from lesion masks. [pause] It learns through **two paths**, paired and self-supervised, with **anatomy-aware conditioning**: a patch-graph neural network and global attention, injected via FiLM. CarDiff reaches the **best FID and LPIPS** of four baselines — the most realistic synthesis.

Slide 3 • Potential Impact

0:40–1:00 • $\sim$ 50 w

So why does it matter? [pause] Using CarDiff images to augment training **five-fold** improves caries segmentation across **seven architectures** — up to **six-and-a-half percent** higher Dice, with the biggest gains on the **rare early lesions**. It's a concrete step toward **trustworthy, AI-assisted diagnosis** when annotated data is scarce. [pause] Thank you.

Delivery tips

- Breathe and **pause** at each [pause] — it resets the clock and adds emphasis.
- Land the three numbers clearly: **best FID/LPIPS**, **5 $\times$** , **+6.5% Dice**.
- Keep energy up on Slide 1 (the hook) and slow slightly on “Thank you.”
- If you run long, drop “superficial and medium” on Slide 1 — saves $\sim$ 2 s.